

Aayush Patel

apatel3088@gatech.edu | +1 (770) 862-5673 | linkedin.com/in/aysh-ptl | github.com/AayushP-418

Full-stack software engineer at Georgia Tech — REST API design, containerized deployment, and AI-integrated product development.

EDUCATION

Georgia Institute of Technology — Atlanta, GA

Expected May 2028

B.S. Computer Science — Systems & Architecture | Minor: FinTech | GPA: 3.71/4.00

Coursework: Data Structures & Algorithms, Objects and Design (CS 2340), Computer Organization & Programming, Machine Learning

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, SQL, C, C++, HTML

Frameworks & Tools: React, Next.js, Node.js, Express, Flask, FastAPI, REST APIs, WebSockets, PostgreSQL, MongoDB, Firebase Auth, Firestore, Docker, Google Cloud Platform (GCP), Linux/Unix, Pydantic, Tailwind CSS, pgvector, OpenCV

Engineering Practices: CI/CD (GitHub Actions), System Design, API Design & Documentation, Database Optimization, Real-Time Streaming (WebSockets, SSE), Connection Pooling, Agile/Scrum, Jira, Git, Unit Testing (pytest, Jest)

WORK EXPERIENCE

Georgia Institute of Technology

May 2026 – Aug 2026

Teaching Assistant, CS 2340: Objects & Design — Atlanta, GA

- Developed course materials on OOP, design patterns, SOLID principles, and Agile/Scrum for 200+ students in CS 2340
- Facilitated office hours and technical presentations on project architecture, debugging techniques, and code review practices
- Earned faculty recognition for communicating software design trade-offs across varied skill levels

Big Data Big Impact, Georgia Tech

Jan 2026 – Present

Director of Projects — Atlanta, GA

- Led 11 cross-functional engineering teams (100+ students) across full-stack, data, and ML tracks
- Standardized the Git workflow with PR review gates; documented REST API interface contracts, enforced via CI/CD
- Drove 6 projects from initial scoping to deployment; presented final MVPs to a Microsoft industry panel

Georgia Tech VIP — EMADE Lab (Advisor: Dr. Jason Zutty)

Aug 2025 – May 2026

Undergraduate Researcher — Evolutionary AutoML & LLM-Guided Synthesis — Atlanta, GA

- Built a modular Python pipeline with 4 pluggable LLM provider integrations for program synthesis and fitness scoring
- Improved synthesis solution quality by 14% on TSP, symbolic regression, and portfolio optimization
- Designed island-migration scheduler for parallel execution; achieved 5.8× speedup over single-island baseline

Mascot Forge Pvt Ltd (automotive-parts manufacturer, India)

May 2025 – Aug 2025

Data Engineering Intern

- Engineered REST API inventory service integrated with PostgreSQL analytics and data pipelines for real-time stock tracking
- Built operational dashboards tracking forging-cycle KPIs for operations teams; reduced reporting latency by 1.3 hours
- Deployed demand-forecasting module to production via REST API integration raising forging-cycle efficiency by 11%

PROJECTS

CodeProctor — Proctored Code Assessment Platform

React | Node.js/Express | PostgreSQL | WebSockets | Firebase

- Built full-stack assessment platform: React frontend, Node.js/Express API, Judge0 sandbox supporting 12 languages
- Cut query latency 28% with node-postgres pooling; WebSocket layer broadcasts live proctoring flags to all active sessions
- Secured with JWT auth, rate limiting, sandboxed execution; core flows validated with Jest + pytest

Research Paper RAG Platform

FastAPI | pgvector | Next.js | Docker | Claude API

- Built FastAPI REST ingestion service with page-aware PDF chunking and Docker Compose deployment for research papers
- Implemented security model: magic-byte validation, secrets isolation, and Pydantic schema contracts across all API endpoints
- Engineered a Pydantic-validated structured extraction with field-level citations and weak-evidence flagging

TerraTrends — Economic Forecasting Platform

Flask | MongoDB | React/Vite | deck.gl | GCP

- Led a team of 8 engineers to architect a Flask + MongoDB + React/Vite full-stack county level economic forecasting app
- Automated ETL ingestion of 15+ public economic and demographic datasets via scheduled cron jobs
- Developed a dynamic deck.gl + Google Maps API dashboard rendering 3,000+ county data points with pagination

ThirdEye — Assistive Vision System

Flask | WebSockets | YOLOv5 | MiDaS | OCR | Gemini API

- At Hackalytics 2025, built Flask/WebSocket backend orchestrating real-time multi-model AI inference
- Integrated Gemini API for contextual scene narration and word replacement via a Chrome extension aiding navigation
- Assembled YOLOv5 detection, MiDaS depth estimation, OCR, and emotion recognition in the backend via parallel processing